



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE

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**2026-2027
LAB RECORD**

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?
Business Intelligence tools help convert raw data into meaningful visual reports and dashboards. They make it easier to identify trends, compare performance, and support better business decisions through interactive data analysis.
2. List any three common data sources that Tableau and Power BI can connect to.
The three common data sources that Tableau and Power BI can connect to.
 - Microsoft Excel
 - CSV (Comma Separated Values) files
 - SQL Database
3. What is a dashboard in BI, and how is it different from a single chart?
A dashboard is a collection of multiple visualizations displayed on one screen to provide a complete overview of the data. A single chart shows only one aspect of the data, whereas a dashboard combines different charts, filters, and KPIs for better analysis.
4. Define the following terms:
 - a) Visualization
Visualization is the graphical representation of data using charts, graphs, maps, or tables to make information easier to understand.
 - b) KPI
A Key Performance Indicator (KPI) is a measurable value that indicates how effectively a business or process is achieving its objectives.
 - c) Data Filter
A data filter is a feature that allows users to display only the required portion of the dataset based on selected conditions.
5. Why is it important to clean and transform data before visualization?
Cleaning and transforming data removes errors, duplicate records, and missing values while organizing the data into a suitable format. This improves the accuracy and reliability of visualizations and ensures meaningful analysis.
6. What kind of chart would you use to analyze stock price trends over time?
A line chart is the most suitable choice because it clearly shows how stock prices change over time and helps identify trends and fluctuations.
7. Explain the role of slicers in Power BI and filters in Tableau.
Slicers in Power BI and filters in Tableau allow users to interact with the data by selecting specific values, such as stock names or dates. This helps focus the analysis on relevant information without changing the original dataset.
8. What does a calculated field or measure do in BI tools?
A calculated field or measure performs custom calculations using existing data. It helps create new values, such as profit, average sales, or prediction error, for deeper analysis.

9. What are the advantages of using interactive dashboards for financial data analysis?
Interactive dashboards allow users to explore financial data quickly, compare different metrics, monitor performance in real time, and identify patterns or unusual trends that support informed decision-making.
10. Which function would you use to find the absolute difference between predicted and actual stock prices?
The **ABSQ** function is used to calculate the absolute difference between predicted and actual stock prices.

IN-LAB EXERCISE:

Objectie:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

Tools Required:

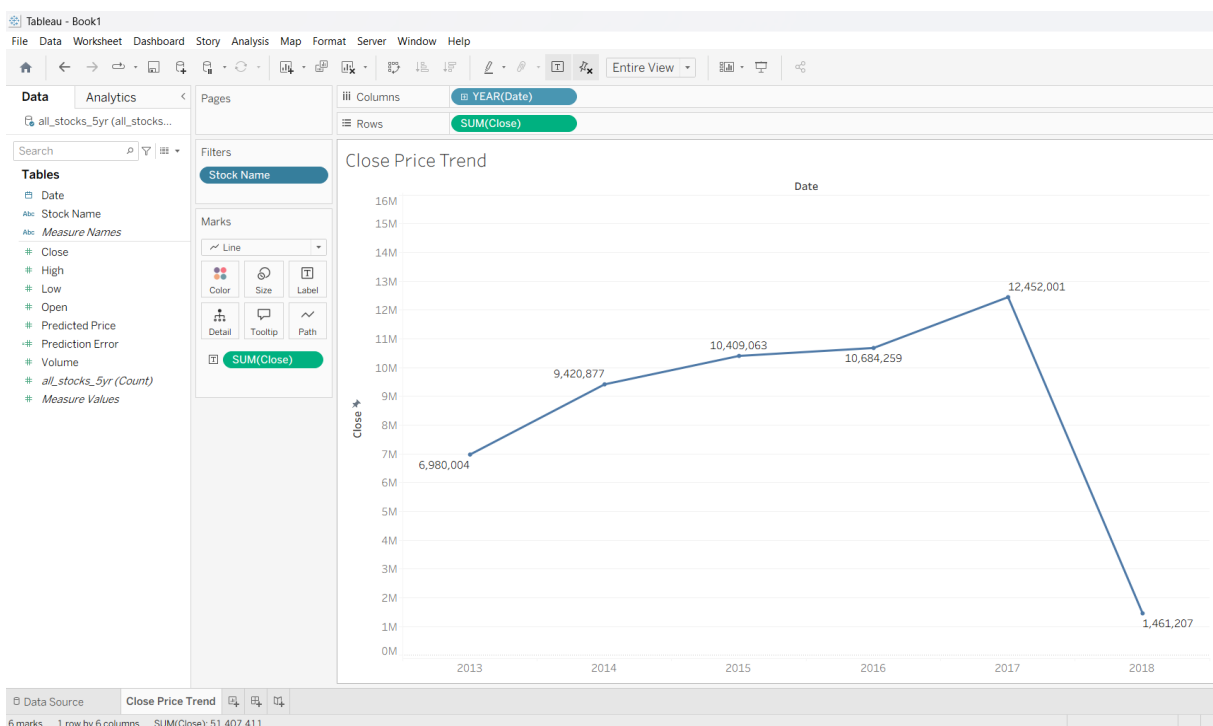
- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

Exercise Steps:

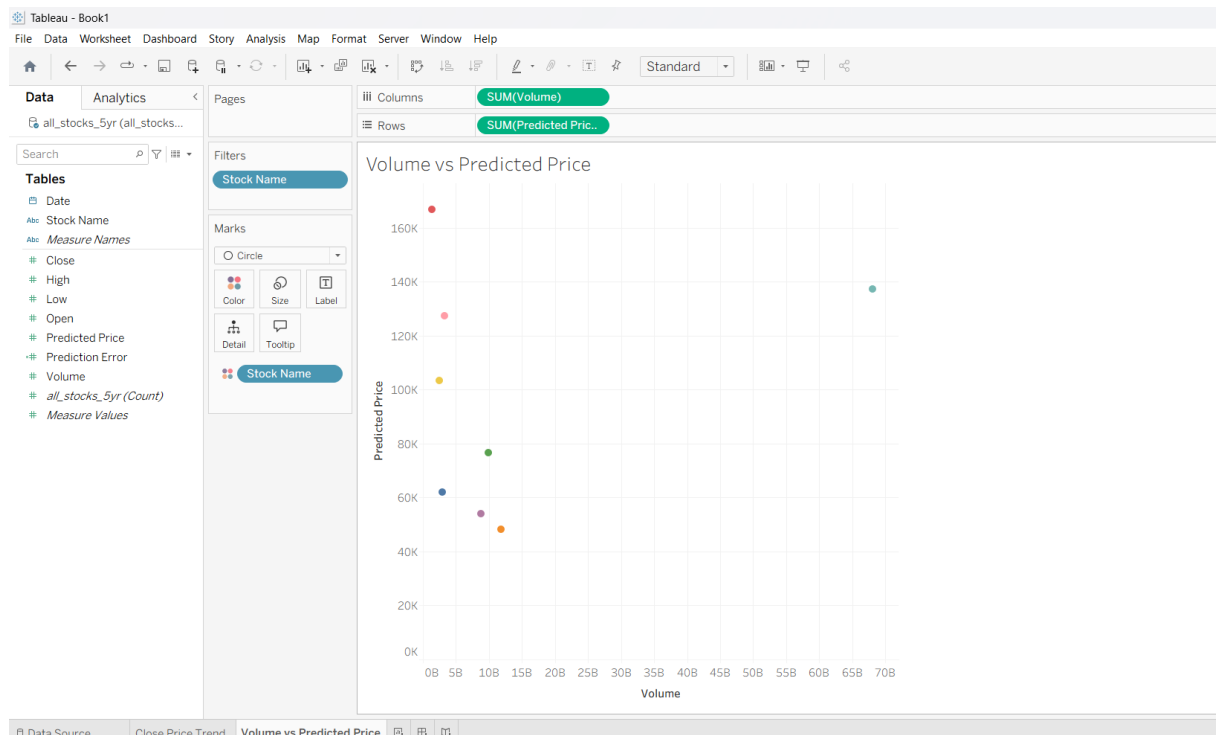
Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:**
 - Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time.
 - **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:**
 - Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

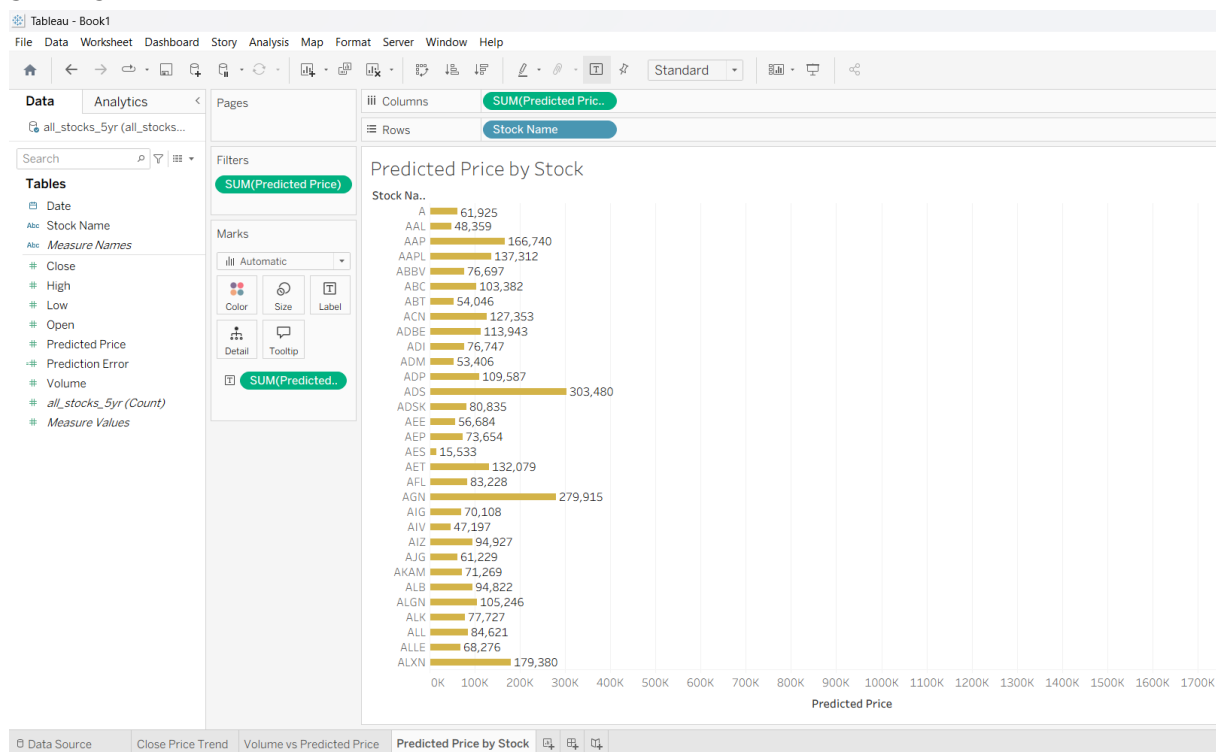
STEP : 1 UPLOADING THE DATASET



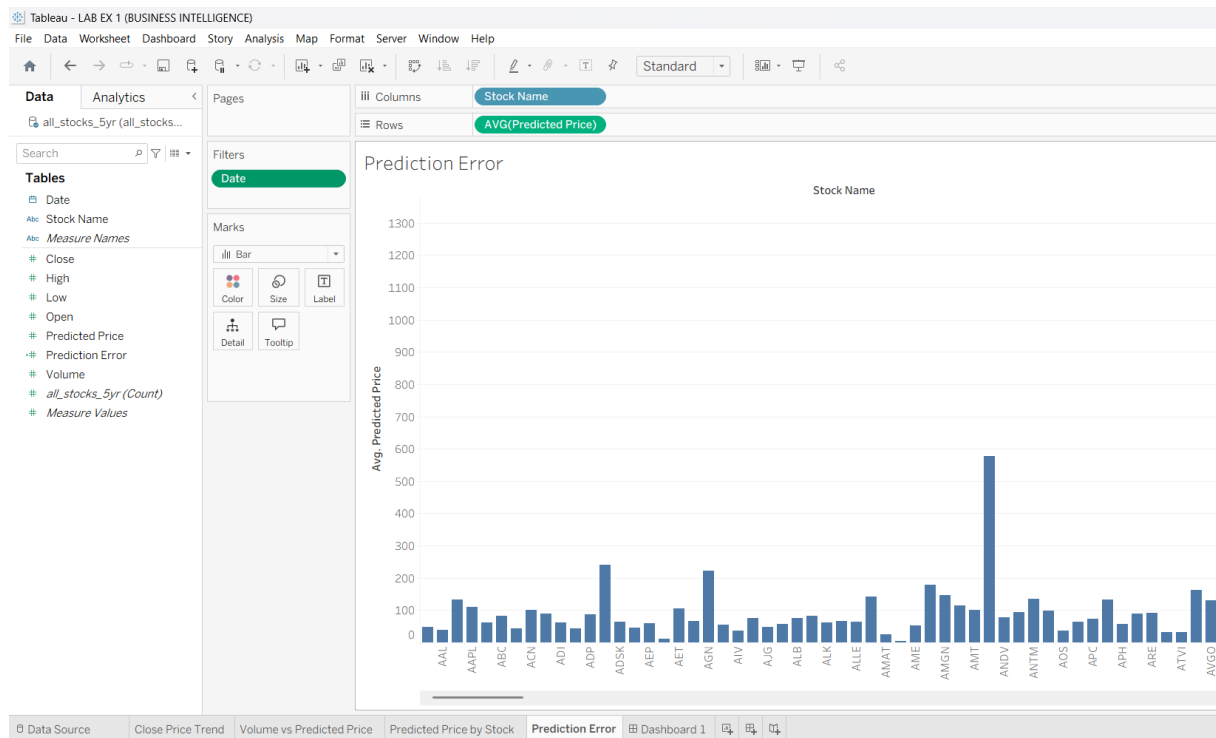
STEP : 2



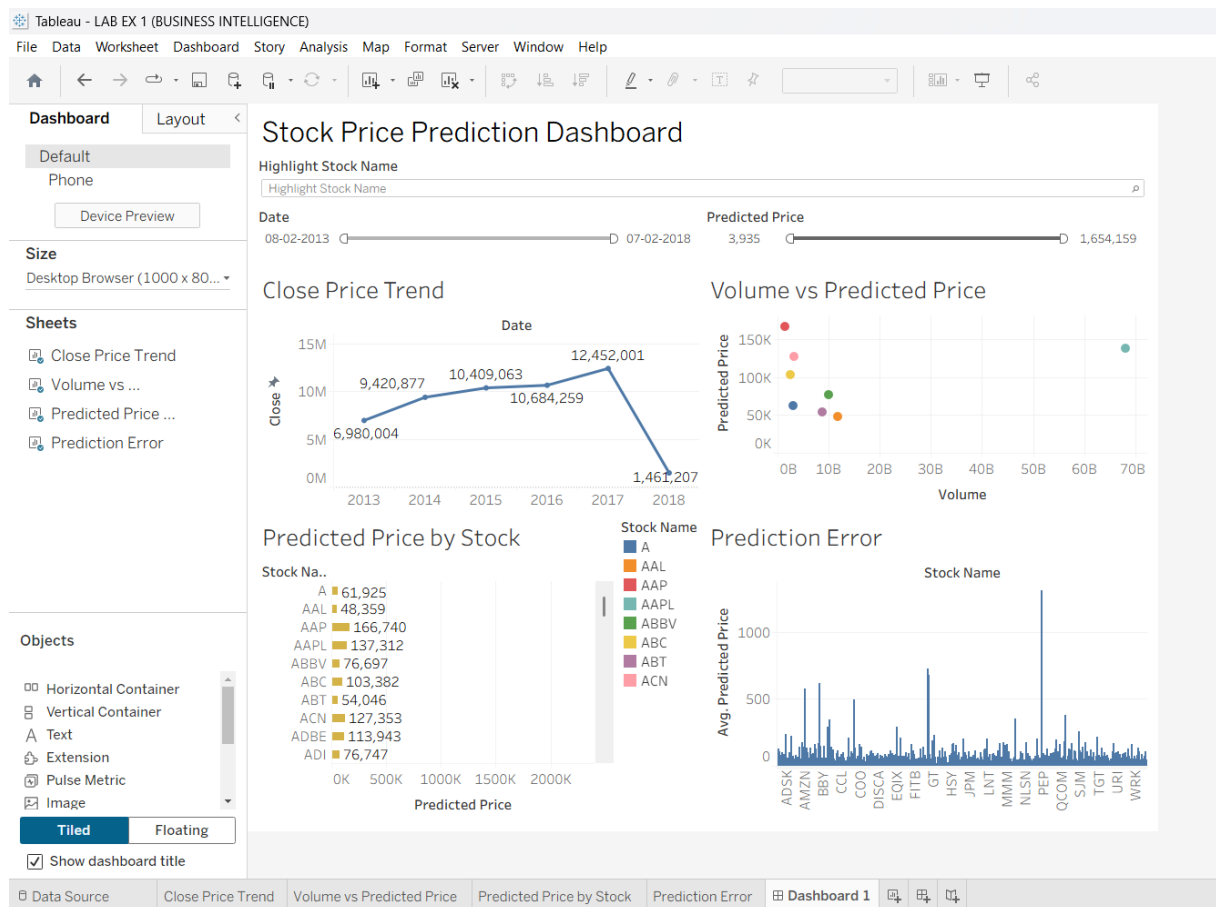
STEP : 3



STEP : 4



STEP : 5



STEP : 6 FINAL DASHBOARD

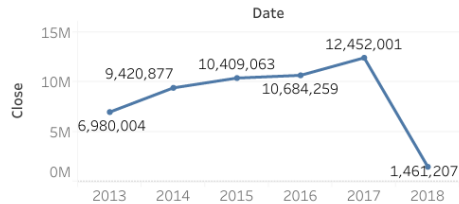
Stock Price Prediction Dashboard

Highlight Stock Name
No items highlighted

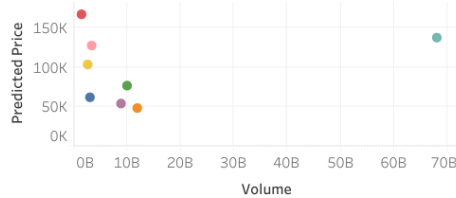
Date
08-02-2013 to 07-02-2018

Predicted Price
To Null

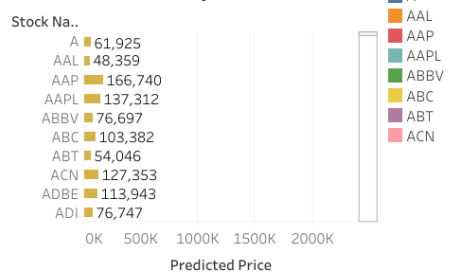
Close Price Trend



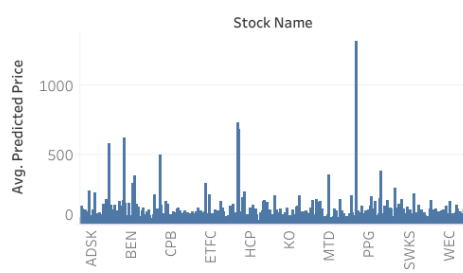
Volume vs Predicted Price



Predicted Price by Stock



Prediction Error



Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:**
 - o Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - o **Line Chart:** Trend of Close Price over time.
 - o **Bar Chart:** Compare Predicted Price vs. Close Price.
 - o **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - o Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - o Example: Prediction Error = $ABS(\text{Predicted} - \text{Close})$
6. **Build a Report Page:**
 - o Arrange visuals in a single interactive page.

STEP : 1

BI LAB EX 1

Home
Transform
Add Column
View
Tools
Help

Close & Apply
Close

New Source
New Query

Recent Sources
New Query

Enter Data
New Query

Data source settings
Data Sources

Manage Parameters
Parameters

Export query results
Output Data

Refresh Preview
Query

Advanced Editor
Manage
Query

Choose Columns
Manage Columns

Remove Columns
Manage Columns

Keep Rows
Reduce Rows

Remove Rows
Reduce Rows

Sort

Split Column
By

Group By

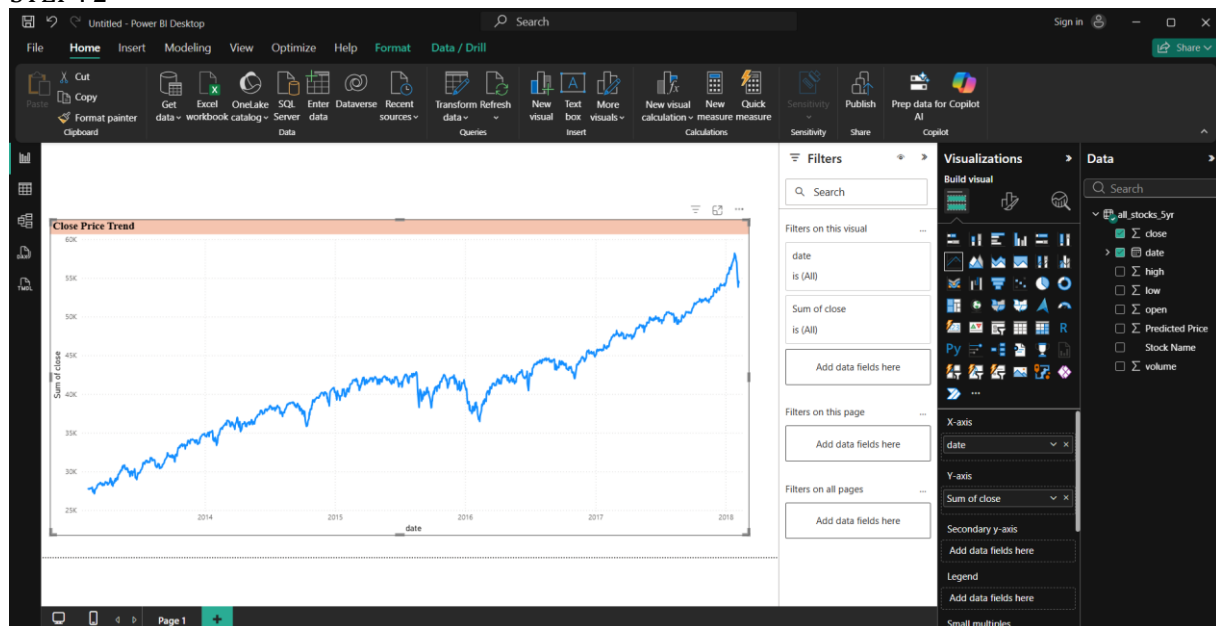
Queries [1]

= Table.TransformColumnTypes("#Promoted Headers",{{"date", type date}, {"open", type number}, {"high", type number}})

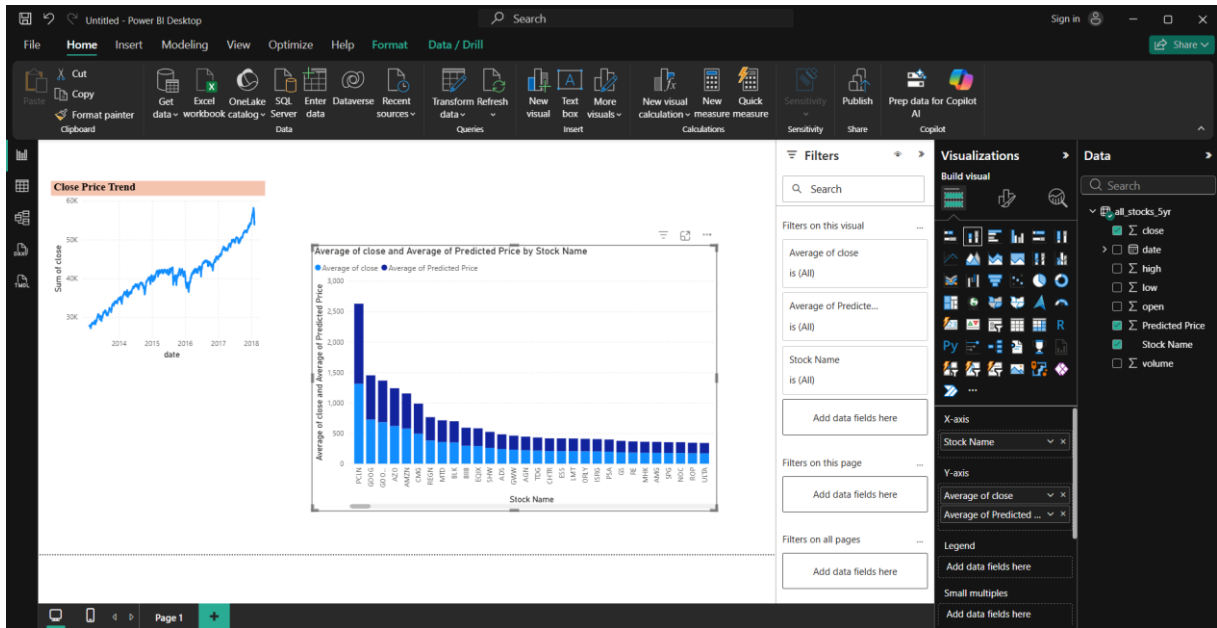
all_stocks_5yr

	date	1.2 open	1.2 high	1.2 low	1.2 close	1.2 volume
1	08-02-2013	15.07	15.12	14.63	14.75	
2	11-02-2013	14.89	15.01	14.26	14.46	
3	12-02-2013	14.45	14.51	14.1	14.27	
4	13-02-2013	14.3	14.94	14.25	14.66	
5	14-02-2013	14.94	14.96	13.16	13.99	
6	15-02-2013	13.93	14.61	13.93	14.5	
7	19-02-2013	14.33	14.56	14.08	14.26	
8	20-02-2013	14.17	14.26	13.15	13.33	
9	21-02-2013	13.62	13.95	12.9	13.37	
10	22-02-2013	13.57	13.6	13.21	13.57	
11	25-02-2013	13.6	13.76	13	13.02	
12	26-02-2013	13.14	13.42	12.7	13.26	
13	27-02-2013	13.28	13.62	13.18	13.41	
14	28-02-2013	13.49	13.63	13.39	13.43	
15	01-03-2013	13.37	13.95	13.32	13.61	
16	04-03-2013	13.5	14.07	13.47	13.9	
17	05-03-2013	14.01	14.05	13.71	14.05	
18	06-03-2013	14.52	14.68	14.25	14.57	

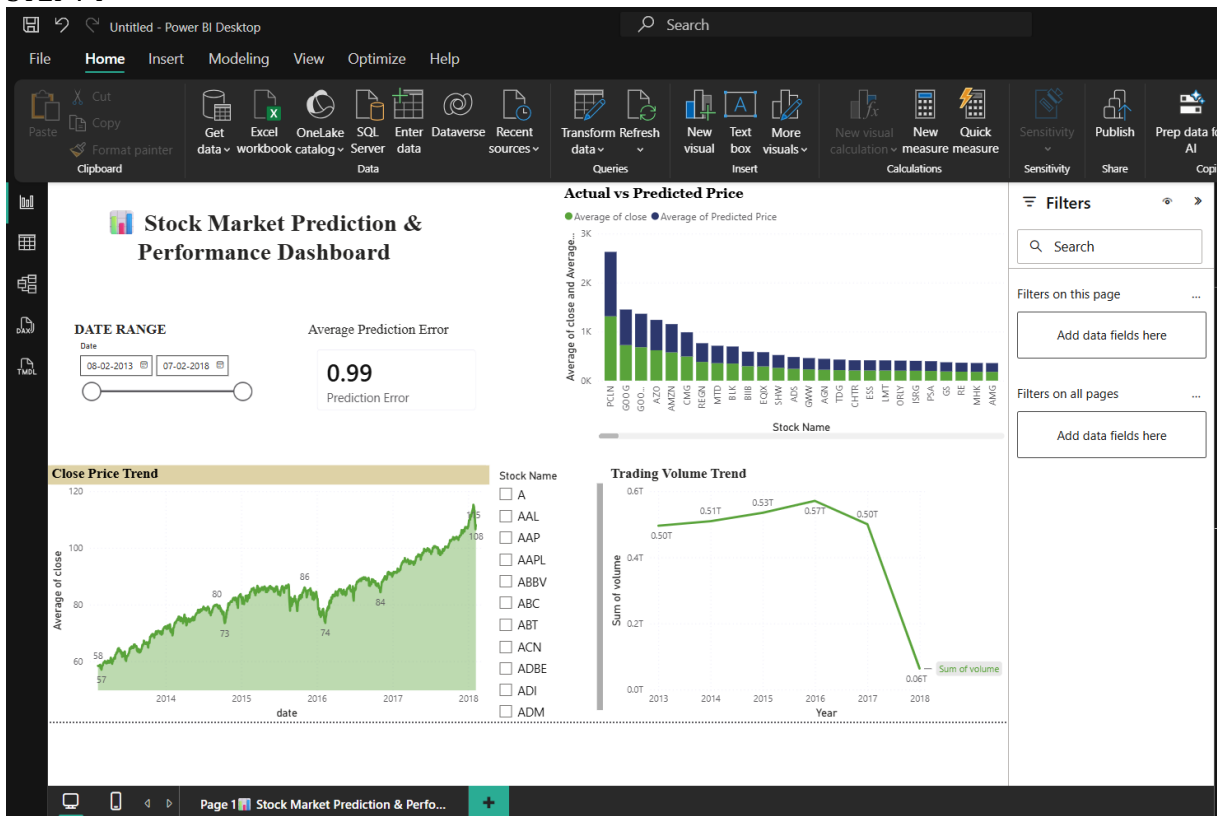
STEP : 2



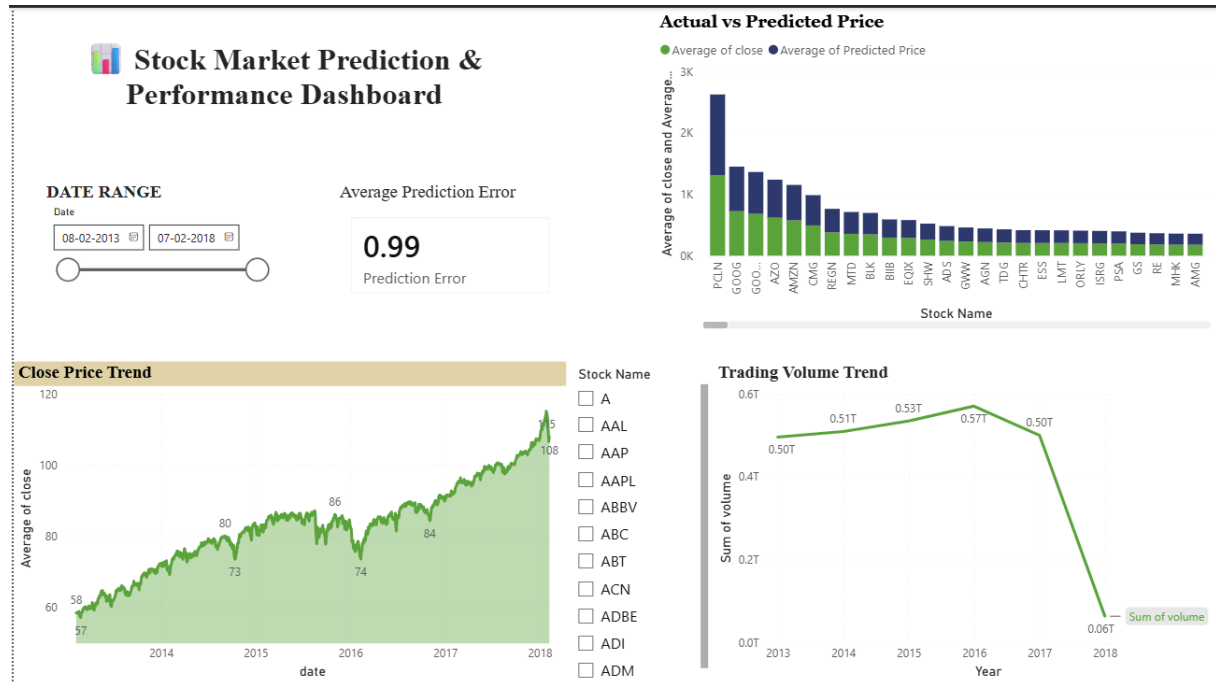
STEP : 3



STEP : 4



STEP : 5 FINAL DASHBOARD



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?

The Tableau interface was simple and organized. I mainly worked with the Data Pane to access the dataset, the Sheets area to create charts, the Marks Card to customize visualizations, and the Dashboard section to combine multiple charts into a single interactive view.

2. How did you import the dataset into Tableau and Power BI?

In Tableau, I selected **Connect** → **Text File**, chose the CSV dataset, and loaded it into the workspace. In Power BI, I clicked **Get Data**, selected the CSV file, previewed the data, and loaded it into the report.

3. What types of visualizations did you create in both tools? List at least two per tool.

Tableau:

- Line Chart showing Close Price over time.
- Scatter Plot displaying Volume versus Predicted Price.
- Highlight Table for Predicted Price by Stock.

Power BI:

- Line Chart showing stock price trends.
- Bar Chart comparing Predicted Price and Close Price.
- KPI Card displaying the average prediction error.

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

The prediction error was calculated by finding the absolute difference between the predicted stock price and the actual closing price using the formula

ABS(Predicted Price – Close Price). It was displayed as a KPI card and also used in charts for comparison.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

I applied filters based on **Stock Name** and **Date**. These helped me focus on individual stocks and specific time periods, making it easier to observe price movements, trading volume, and prediction accuracy.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?

In Tableau, creating a line chart mainly involved dragging fields onto the Rows and Columns shelves. In Power BI, I first selected the Line Chart visual and then assigned fields to the Axis and Values sections. Tableau felt more drag-and-drop oriented, while Power BI relied more on configuring visual properties.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

I arranged the charts neatly so that the line chart was placed at the top for trend analysis, followed by comparison charts and the KPI card. I also included filters for Stock Name and Date to make the dashboard interactive and easy to use.

8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?

I found **Power BI** slightly more user-friendly because its interface is familiar to users who have worked with Microsoft products. It was easier to create reports and KPIs, while Tableau provided greater flexibility for designing advanced visualizations.

9. What challenges did you face while creating visualizations or dashboards?
Initially, I found it difficult to choose the correct fields for each visualization and arrange multiple charts effectively. After understanding the interface and using filters correctly, creating the dashboard became much easier.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The concepts learned in this lab can be used to analyze stock market trends, monitor investment performance, compare predicted and actual prices, and build interactive dashboards that help financial analysts and investors make informed decisions based on data.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		